

Task 11 – Comparative Analysis of Classification Models

Logistic Regression vs. Random Forest vs. SVM on the Titanic Dataset

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Part A – Dataset Selection & Preparation

Dataset Description

The Titanic Passenger Survival dataset (891 records, 12 columns) is a widely used public benchmark for binary classification. It records passenger demographics and travel details from the RMS Titanic, with the objective of predicting survival.

Target Variable

Survived — binary (0 = Did not survive, 1 = Survived).

Features Used

- Pclass — Passenger travel class (1st, 2nd, 3rd)
- Sex — Gender (encoded numerically)
- Age — Age in years (missing values imputed with median)
- SibSp — Number of siblings/spouses aboard
- Parch — Number of parents/children aboard
- Fare — Ticket fare
- Embarked — Port of embarkation (encoded numerically)

Columns PassengerId, Name, Ticket, and Cabin were dropped as identifiers or due to excessive missing data.

Preprocessing Steps

- Imputed missing Age with median; missing Embarked with mode
- Label-encoded categorical columns Sex and Embarked
- Standardized numeric features using StandardScaler (for Logistic Regression and SVM)
- Split data 80% training / 20% testing, stratified by target class (random_state=42)

Part B – Model Development

All three models were trained on identical train/test splits to ensure a fair comparison.

1. Logistic Regression

A linear model estimating the probability of survival using a sigmoid function. Trained on scaled features with max_iter=1000.

2. Random Forest

An ensemble of 200 decision trees using bagging and majority voting. Trained on unscaled features, since tree-based models are scale-invariant.

3. Support Vector Machine (SVM)

An RBF-kernel SVM that finds the optimal separating hyperplane in a higher-dimensional feature space. Trained on scaled features.

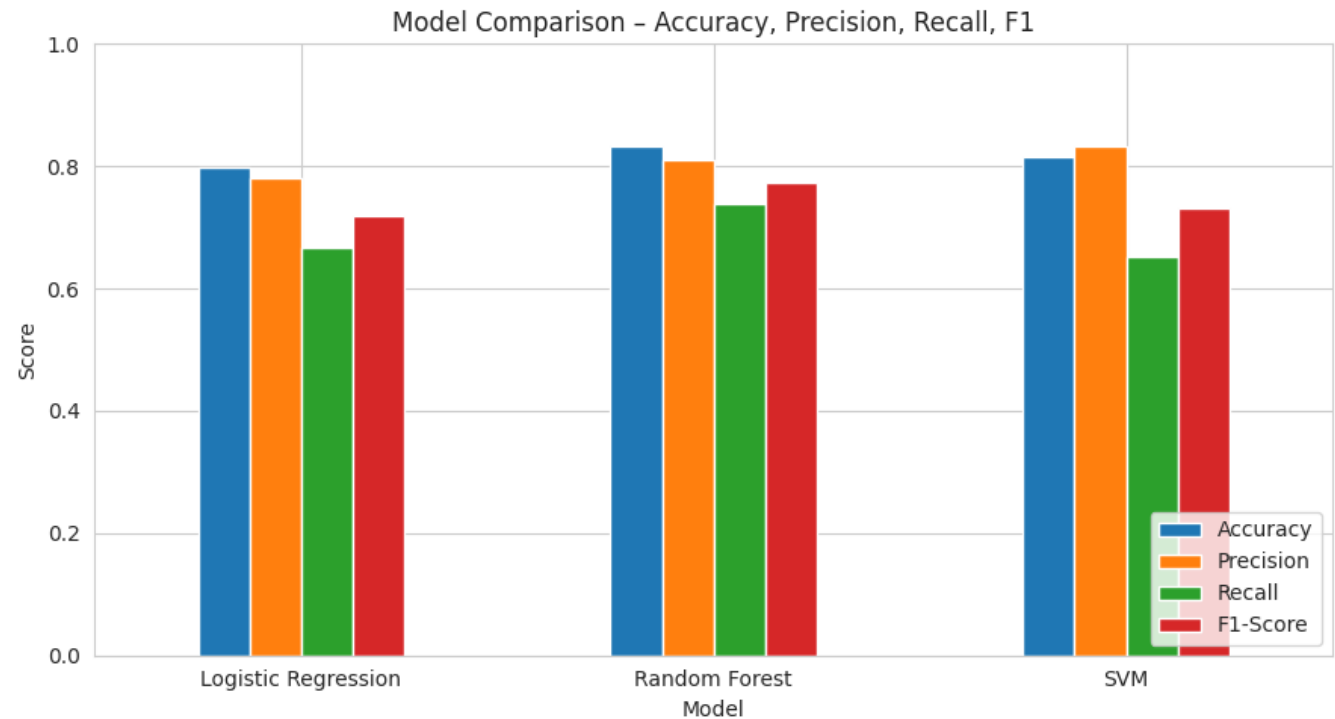
Part C – Model Evaluation & Comparison

Each model was evaluated on the same held-out test set (20% of data, 179 samples) using Accuracy, Precision, Recall, and F1-Score.

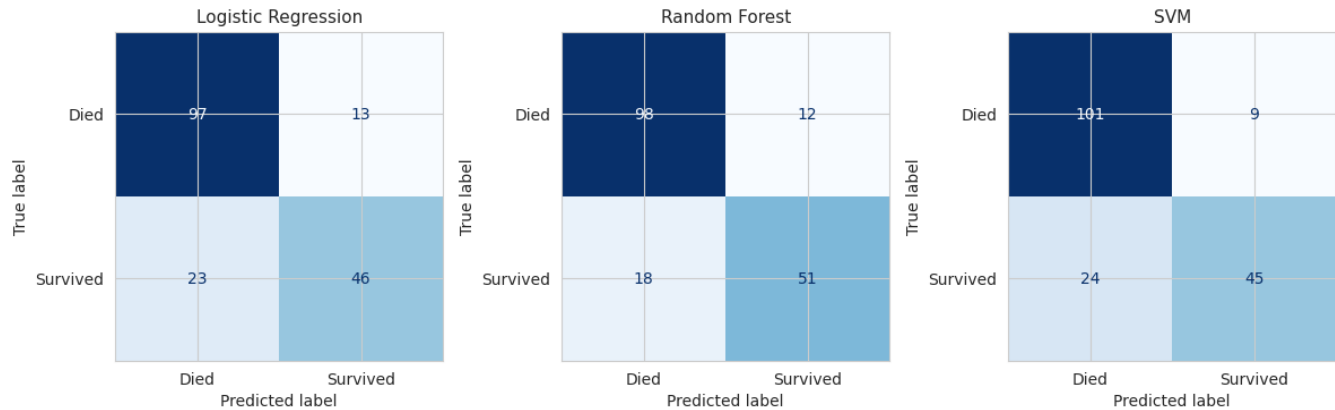
Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.799	0.780	0.667	0.719
Random Forest	0.832	0.810	0.739	0.773
SVM	0.816	0.833	0.652	0.732

Metric Comparison Chart



Confusion Matrices



Strengths & Weaknesses

Model	Strengths	Weaknesses
Logistic Regression	Fast, interpretable, low variance, clear feature impact via coefficients	Assumes linear decision boundary; weaker on complex, non-linear patterns
Random Forest	Captures non-linear relationships, robust to outliers, gives feature importance	Less interpretable; risk of overfitting with excessive depth/trees
SVM	Effective in high-dimensional space; strong margin-based separation	Sensitive to feature scaling and kernel/parameter choice; slower on large data

Part D – Recommendation & Conclusion

Based on F1-Score — which balances precision and recall — Random Forest is the best-performing model, achieving the highest Accuracy (0.832), Recall (0.739), and F1-Score (0.773) among the three.

Random Forest outperforms Logistic Regression because it captures non-linear interactions between features such as Sex, Pclass, and Fare, which a linear decision boundary cannot fully model. It also edges out SVM, whose performance is more sensitive to feature scaling and kernel choice on this moderately sized dataset.

Recommendation

Random Forest is recommended for deployment given its balanced performance across all metrics. If model interpretability is a priority — for example, explaining survival factors to non-technical stakeholders — Logistic Regression remains a strong, more transparent alternative despite its slightly lower F1-Score.